State inputs to the ECEC simulator, 10/14

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Since Oct. 9, I made changes to the ECEC simulator's state inputs and reran calibration for 50 states, DC, and NYC. The base year is 2019, and demand covers ages 0–4. NY includes NYC, whose separate run is excluded from national totals. All numbers in this memo come from the run on the evening of October 13th.

- **Care labels and training.** The care model learns from NSECE. I now count regular care that didn't report any hours during the survey-week, use care type to fill missing regular-care answers, and exclude unknown-type arrangements. These changes raise the weighted share of children ages 0–4 in regular care from 45.5% to 55.0%, matching NSECE's own measure. I fill uncertain hours, paid status, and price tier with random draws. To reduce dependence on the random seed, I make the draws 50 times and train on all 50 versions together. Two-parent families now use mothers' hours, matching ACS (larger discussion).
- **Income ranks.** Ranks stay national, but now use income adjusted for local prices instead of nominal income, so a family's rank reflects what its income buys where it lives. As in the simulator, the price model ranks tax-unit AGI and the care model ranks household income. Both keep the price adjustment.
- **High-low threshold.** In state runs, the threshold between high-price and low-price centers now depends on region and area urbanicity as well as child age. National runs keep the original threshold: one national median for each child age band.
- **Participation.** Every geography now uses one five-hour factor, 1.099, to extend NSCH's ten-hour participation rate to regular care starting at five hours.
- **Staffing.** Required staff per child now uses 2017 licensing rules instead of 2014 rules.
- **Household payments.** The results in this memo use an interim method: each family draws its price multiplier at random from what NSECE households actually paid, relative to the national price of that care type. Before, a forest predicted it. I am restoring the original price forest for all runs, with its original settings.

All 104 equilibrium solves for 2026–27 converged. Base-year child counts by care type are within 2.4% of target.

Target construction

1. I multiply the child population by NSCH participation and the five-hour factor. NYC uses the NY participation rate.
2. I subtract unpaid-center children from the estimated number in regular care, using pre-K, Head Start, and Early Head Start enrollment with overlaps removed. Predicted care shares split the remaining children across other types.
3. I assign 500 annual hours per part-time child and 2,000 per full-time child.

1. Care model

1.1 Training labels and imputation

The model predicts each child's care type and hours band. NSECE asks about each provider that cares for the child. Three answers matter: whether the household uses the provider regularly, the type of care, and the hours of care in the survey week.

NSECE leaves some answers missing or unusable. I handle each gap with a fixed rule or a random draw. These changes raise the share of children in regular care from 45.5% to 55.0%, which equals NSECE's own measure for ages 0–4: any provider of type 1 to 5. The percentages below are shares of all weighted NSECE children ages 0–4.

1. **Zero or missing hours in the survey week, filled by a random draw.** A regular provider can have zero or missing hours because the child used it regularly but not that week. It now counts as care, and I draw the hours from regular providers with the same type of care and child age, and in state runs the same region, using survey weights. This moves 6.7% of children from parent only to care.
2. **Missing regular-care answer, handled by a fixed rule.** Some households refused the question, or the interviewer could not determine the answer. These children now stay in the training data, and the care counts when the type is 1 to 5. They represent 4.0%: 3.7% in care and 0.3% parent only.
3. **Irregular or unknown type of care, handled by a fixed rule.** Care of irregular or unknown type no longer counts, even when the household calls it regular. This moves 0.6% of children from other unpaid care to parent only.
4. **Unknown payment amount, filled by a random draw.** Some households refused the weekly payment, did not know it, or skipped the payment section. If they still said whether they pay the provider anything, I use that answer. Otherwise I draw paid or unpaid from households with a known payment, the same kind of provider, and the same child age, and in state runs the same region. I assume the two groups resemble each other.
5. **Paid center with no hourly price, filled by a random draw.** I sort paid centers into high-price and low-price by the hourly price paid. A center with no usable price cannot be sorted, so I assign a tier at random. I use the high-price share among paid centers with a known price and the same child age. State runs also match on region and area urbanicity, where enough centers exist.

When a child uses several care types, I choose one in this order: high-price center, low-price center, unpaid center, paid home, other paid, unpaid home, other unpaid. I sum hours in that type; fewer than 20 weekly hours counts as part-time.

Items 1, 4, and 5 use random draws. I create 50 label versions, each with one-fiftieth of each child's survey weight, keeping siblings together. The table shows ranges across random seeds for weighted children ages 0–4: 20 seeds for one draw and five for 50 versions.

	One draw	50 versions
High-price share of paid-center children	50.2–53.6%	51.9–52.1%
Part-time share of children in care	25.1–26.5%	26.0–26.1%
Five-hour factor (section 2)	1.094–1.105	1.099–1.100

I count children before dropping unlabeled records, avoiding false one-child families. Two-parent families now use mothers' hours, matching ACS.

1.2 Income rank

The original model ranks families across the country by nominal income. I keep the national ranks but divide income by the local price level first: income $\div$ (RPP/100). RPP is the 2019 BEA regional price parity, with the US at 100. NSECE uses regional price levels. NYC's run uses the metro price level.

The original price model ranks each family by its baseline AGI, the sum over both parents. The care model ranks household income. For now I match the simulator: the price model ranks AGI and the care model ranks household income, and both keep the price adjustment. The two models use different income measures, which is an open question in section 6.

1.3 Price tiers

Prices at or above the threshold are high-price; lower prices are low-price. I apply the threshold to posted prices and household payments, which can put the same care in different tiers. Tiers describe relative price, not quality. National runs use one median for each child age band. In state runs, the threshold varies by region, child age, and area urbanicity: high, moderate, or rural.

$$\text{threshold} = \begin{matrix}\text{region's median price,}\\ \text{high urbanicity}\end{matrix} \times \frac{\begin{matrix}\text{national median price,}\\ \text{same urbanicity}\end{matrix}}{\begin{matrix}\text{national median price,}\\ \text{high urbanicity}\end{matrix}}$$

Each median is the weighted median posted price of centers for one age band. Moderate-area median prices are 85–95% of dense-area medians; rural medians are 67–76%, depending on child age. Regional rural samples have only 5–35 centers per age band, so I use national ratios and assume they hold across regions.

2. Participation

The NSCH participation rate counts children in care for 10 or more hours a week. The model counts regular care, which starts at 5 hours. The five-hour factor converts one to the other:

$$\text{five-hour factor} = \frac{\text{NSECE children in regular care}}{\text{NSECE children in care 10 or more hours a week}}$$

Before, I adjusted the national factor using each geography's predicted part-time share. I now use one national factor, 1.099, from corrected NSECE labels averaged over 50 versions.

NHES is a second national child care survey which provides validation. After adjustments to match the model's scope, NHES gives 1.077, about 2% below NSECE.

3. Supply

State wages vary by an OES wage index for each education group. I subtract wage costs from the national price to estimate non-wage costs, such as rent and supplies. Every state uses this same residual.

The national model's staff per child comes from NSECE classroom staff counts. A staffing index scales that national staff need to each state's licensing rules. I make one index for centers and one for homes:

$$\text{staffing index} = \frac{\text{state's required staff per child}}{\text{national average, states weighted by children 0–4}}$$

- The rules come from the ACF 2017 Child Care Licensing Study (ICPSR 37700). Centers use the ratios at five child ages. Homes use one ratio for all children under school age.
- I weight the national paid-home staffing estimates with NSECE provider weights.
- NYC centers use the June 2019 city Health Code. Idaho's missing ratios come from its points rule.

4. Household payment multipliers

$$\text{family price} = \text{state market price} \times \text{multiplier}$$

A multiplier says how far a household's price is from the national price of its care type. A multiplier of 0.5 means the household paid half the national price, and 2 means double. Each family draws one multiplier at random from the NSECE households that pay for its care type. Low-price centers and other paid care use separate draws by income fifth. High-price centers and paid homes use one draw for all incomes. **This is an interim method, and I am replacing it with the original price forest.** Caveats:

- The draw pools households from all regions, so it may overstate the spread within a state.
- Each care type has only about 200 to 600 observations.
- The income groups depend on the income rank, which is AGI in the price model.

5. Calibration

- Annual hours match targets exactly.
- All 52 base-year demand adjustments converged.

5.1 Against posted prices

- A 40-hour week of paid center care costs $227 in the model versus $212 in the 2019 National Database of Childcare Prices (NDCP).
- Across 46 geographies, prices correlate at 0.79; 23 are within 15%. The model exceeds NDCP by 12% at the median, with the largest relative gaps where posted prices are low.
- NDCP combines county median posted prices using child weights. The model combines high-price and low-price center prices using child weights, then multiplies by 40. Subsidies, age mix, hours, and aggregation limit comparability.

6. Remaining issues and next steps

1. **Income ranks.** The simulator is not consistent: the price model ranks AGI and the care model ranks household income. The price forest also trains on NSECE household income and applies to ACS AGI. I match the simulator for now. Open questions: whether both models should use one measure, and how the choice changes the price forest and the care forest. The all-items price level may not match local child care costs, which I will also check.

2. **Urbanicity.** I am dropping the urbanicity variable from the care forest as it left results essentially unchanged. The urbanicity-based tier threshold stays, because rural and moderate-area centers have lower posted prices than dense-area centers. I limit it to state runs by a setting that is off by default.
3. **Matching for random draws.** In state runs, the hours, paid-status, and tier draws all match on region as well as care type and child age. National runs match on care type and child age only. NSECE has no state identifier, so region is the finest donor geography. The cells shrink, so each draw has fewer donors. Where a cell has fewer than 30 donors, the draw falls back to the broader cell.
4. **Price forest.** I am restoring the original forest for all runs, with its original settings and no tuning, as in the simulator. Then I will rerun the state results. This is in progress.
5. **Staffing.** The 2017 licensing limits predate 2019. I will compare them with 2019 rules and observed NSECE staffing.
6. **Posted prices.** Scaling non-wage costs by state price levels raises the base-year correlation with 2019 posted prices from 0.79 to 0.83. In equilibrium, the 2026 correlation rises from 0.77 to 0.80, but DC's paid-center share shift grows from 7.6 to 10.2 percentage points. I will align price definitions before adopting the adjustment.
7. **The 2026 shift.** Paid-center share rises most where low-price center prices grow faster (correlation 0.80). I will find out why.

7. Changes that also affect the national model

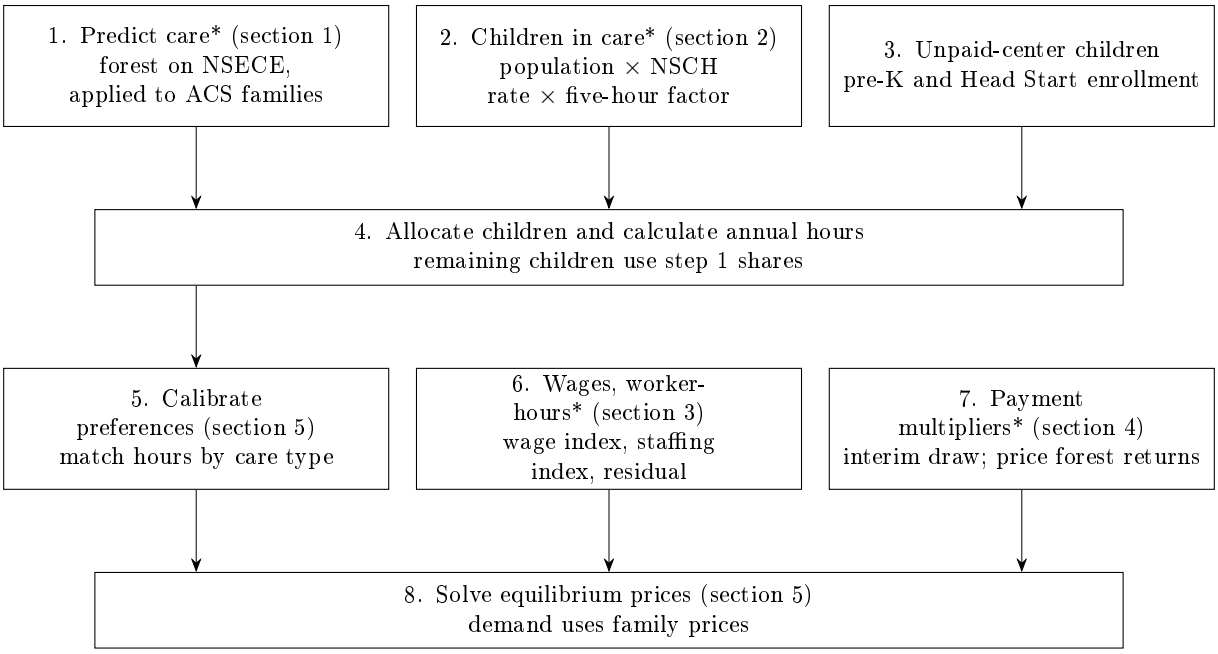
These changes are in code that the national model also runs, on the state-inputs branch. None is in the main branch yet. I have not measured how much a national run would change.

- Label rules and the 50 random draws change the NSECE training labels in every run, so the care forest trains on different data (section 1.1).
- The caregiver rule uses mothers' hours in two-parent families. The forest uses them as a variable, and they also set hourly earnings. Pull request 7 makes the same change on the main branch.
- National income ranks, adjusted for local prices, are the default. They set a forest variable and the payment income group (section 1.2).

Only these inputs are limited to state runs: demand targets, the five-hour factor, state wages, state staffing indexes, state employment files, region matching in the random draws, and the urbanicity-based tier threshold. The first five load only when a run names states. One setting turns on the last two, and state setup runs with it on. With it off, the model matches the main branch on region and urbanicity.

Appendix. How the inputs enter the model

An asterisk marks a step that changed since 9 October.


1. Predict care* (section 1)
forest on NSECE,
applied to ACS families
2. Children in care* (section 2)
population × NSCH
rate × five-hour factor
3. Unpaid-center children
pre-K and Head Start enrollment
4. Allocate children and calculate annual hours
remaining children use step 1 shares
5. Calibrate
preferences (section 5)
match hours by care type
6. Wages, worker-
hours* (section 3)
wage index, staffing
index, residual
7. Payment
multipliers* (section 4)
interim draw; price forest returns
8. Solve equilibrium prices (section 5)
demand uses family prices